2017 Exam Solutions F78PB.

1. (a) (i) $\bar{x} = \frac{1448}{19} = 76.211$

Median $= x_{\left(\frac{19+1}{2}\right)} = x_{(10)} = 78$ (remember to sort!)

$$s = \sqrt{\frac{1}{18}\left[114992 - \frac{1448^2}{19}\right]} = 16.054$$

$Q_1 = x_{(1+18\times0.25)} = x_{(5.5)} = \frac{69+63}{2} = \underline{\underline{66}}$

$Q_3 = x_{(1+18\times0.75)} = x_{(14.5)} = \underline{\underline{86}}$

IQR $= 86 - 66 = \underline{20}$ This measures the middle 50% of the data.

(ii)





The data are skewed to the left (negatively skewed).

(iii) Because the data are skewed, extreme points affect mean and s.d. more than they do median and IQR.

(b) Coefficient of Variation for Everlast $= \frac{2}{50} = 0.04$

" " " for Endurance $= \frac{6}{50} = 0.12$

Since the CV is smaller for Everlast this seems a better choice.

2(a) $\text{Bias}(g(\underline{X})) = E(g(\underline{X}) - \theta)$ 1

$\text{MSE}(g(\underline{X})) = E((g(\underline{X}) - \theta)^2)$ 1

$\text{MSE}(g(\underline{X})) = \text{Bias}^2(g(\underline{X})) + \text{Var}(g(\underline{X}))$ 2.

(b)(i) $\text{Bias}\left(\frac{\bar{X}}{2}\right) = E\left(\frac{\bar{X}}{2} - \theta\right)$

$= \frac{1}{2}E(\bar{X}) - E(\theta)$.

$= \frac{1}{2}E(X) - \theta$

$= \frac{1}{2} \cdot 2\theta - \theta$ 1

$= \underline{\underline{0}}$ so unbiased.

$\text{Var}\left(\frac{\bar{X}}{2}\right) = \frac{1}{4}\text{var}(\bar{X})$

$= \frac{1}{4n}\text{var}(X)$

$= \frac{1}{4n} \cdot 2\theta^2$

$= \underline{\underline{\frac{\theta^2}{2n}}}$ 1

So $\text{MSE}\left(\frac{\bar{X}}{2}\right) = \frac{\theta^2}{2n}$ $(\to 0)$ 1

As $n \to \infty$, MSE $\to 0$ which is good 1

2(b)(ii) $Y = c\bar{X}$

$$\begin{aligned} \text{bias}(Y) &= E(Y-\theta) \\ &= E(c\bar{X}) - \theta \\ &= cE(\bar{X}) - \theta \\ &= cE(X) - \theta \\ &= 2c\theta - \theta \\ &= (2c-1)\theta \end{aligned}$$ 1

$$\begin{aligned} \text{Var}(Y) &= \text{Var}(c\bar{X}) \\ &= c^2 \text{Var}(\bar{X}) \\ &= \frac{c^2}{n} \text{Var}(X) \\ &= \frac{2\theta^2 c^2}{n} \end{aligned}$$ 1

Using part (a), $MSE(Y) = (2c-1)^2\theta^2 + \frac{2c^2\theta^2}{n}$

as required. 2

3 (a) $\int_0^1 (\theta+1)x^\theta \, x \, dx = E(x)$

$= \int_0^1 (\theta+1)x^{(\theta+1)} dx$

$= \left[\frac{(\theta+1)}{(\theta+2)} x^{(\theta+2)}\right]_0^1$

$= \frac{\theta+1}{\theta+2}$

(b) Sample first moment is $\bar{X}$
Population first moment is $\frac{\theta+1}{\theta+2}$

Method of Moments $\Rightarrow$ Equate the two.

$$\bar{X} = \frac{\hat{\theta}+1}{\hat{\theta}+2}$$

$$\Rightarrow \hat{\theta} = \frac{2\bar{X}-1}{1-\bar{X}}$$

(c) Advantage - generally easy to apply (as above).
Disadvantage - in some circumstances results are nonsensical or not available.

(d) $L(\theta) = \prod_{i=1}^{n} (\theta+1) x_i^\theta$

$= (\theta+1)^n \prod_{i=1}^{n} x_i^\theta$

$$l(\theta) = n\log(\theta+1) + \theta \sum \log(x_i)$$

$$\frac{d\ell}{d\theta} = \frac{n}{\theta+1} + \sum \log(x_i)$$

Equate to 0. $\Rightarrow \quad \frac{n}{\hat{\theta}+1} = -\sum \log(x_i)$

$$\Rightarrow \quad \hat{\theta} = \frac{-n}{\sum \log(x_i)} - 1$$

check for max: $\frac{d^2\ell}{d\theta^2} = -\frac{n}{(\theta+1)^2} < 0$ $\Rightarrow$ MAX.

(6)

4. (a) The 95% confidence interval for the ratio $\frac{\sigma_1^2}{\sigma_2^2}$ is

$$\left(\frac{s_1^2 / s_2^2}{F_{7,8,0.025}}, \frac{s_1^2 / s_2^2}{F_{7,8,0.975}} \right)$$

where $F_{7,8,0.025} = 4.529$ and $F_{7,8,0.975} = \frac{1}{F_{8,7,0.025}} = \frac{1}{4.899} = 0.2041$ [1]

$$\left(\frac{101.17/94.73}{4.529}, \frac{101.17/94.73}{0.2041} \right)$$ [1]

$(0.2358, 7.8775)$ [1]

Since 1 lies inside this confidence interval, we can conclude that the two population variances are equal at α=0.05. [1]

(b)

(i) It is a function of the sample and $\mu_1 - \mu_2$, it has a known distribution (t) and it is monotonic.

[2]

(ii) From part (a) above, equal population variances can be assumed so pivotal quantity is valid.

$$s_p = \sqrt{\frac{7(101.17) + 8(94.73)}{8+9-2}} = 9.8861$$

$$\text{CI} = 2.3 +/- \ 2.131^* \ 9.8861\sqrt{\frac{1}{8} + \frac{1}{9}}$$

~~=2.3 +/- 4.8~~

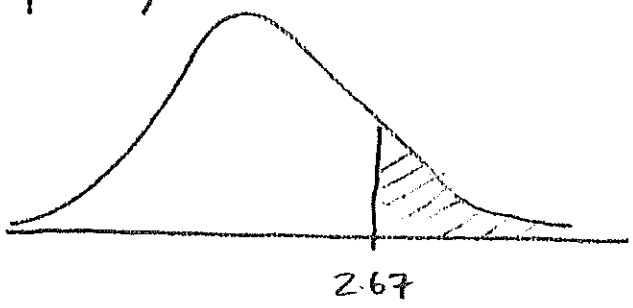
~~=(-2.5, 7.1)~~ $= (-7.937, 12.537)$

We can conclude that the filtering device does not change the percentage of impurity significantly at α=0.05 (since 0 is included in the CI).

[3]

5. (a) $H_0 : \mu = 8$

$H_1 : \mu > 8$ 1



Calculate t statistic $t = \dfrac{\bar{x} - \mu}{s/\sqrt{n}}$

$= \dfrac{9.94 - 8}{2.91/\sqrt{16}}$

$= 2.67$ 1

Calculate p-value (15 df)

$t = 2.6$	gives	0.9900
$t = 2.7$	gives	0.9918

$t = 2.67 \Rightarrow$ Area $= 0.9900 + \frac{7}{10}(0.9918 - 0.9900)$

$= 0.9913$

So p-value is ~~0.0076~~ 0.0087 1

Since this is quite small we have strong evidence against H_0. There is evidence that patients have the virus for longer than 8 days. 1

Data need to be normal : No evidence against this on dotplot.



(b) Type I error is when H_0 is rejected when it is true. 1

Type II error is the probability to accept H_0 when it is false 1

6. (a)

$$S_{xx} = \sum x_i^2 - \frac{1}{10}\left(\sum x_i\right)^2 = 32.655$$

$$S_{yy} = \sum y_i^2 - \frac{1}{10}\left(\sum y_i\right)^2 = 162.10$$

$$S_{xy} = \sum x_i y_i - \frac{1}{10}\left(\sum x_i\right)\left(\sum y_i\right) = 64.25 \qquad \text{[0.5 Each]}$$

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = 1.967$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = -11.445$$

The fitted line is $\hat{y} = 1.967x - 11.445$. [1.5]

(b) (i) The scatter plot shows a positive linear relationship between blood flow and intraocular pressure. [1]

(ii) From the residuals plot, the residuals are randomly scattered. It seems to indicate that the residuals are independent and follow normal distribution with mean 0 and constant variance. The linear model is adequate for these data. [2]

(c) The 95% confidence interval for β is

$$\hat{\beta} \pm t_{8,0.025}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \qquad \text{where} \quad \hat{\sigma}^2 = \frac{1}{10-2}\left[S_{yy} - \frac{S_{xy}^2}{S_{xx}}\right] = 4.461, \; t_{8,0.025} = 2.306 \qquad \text{[1]}$$

$$1.967 \pm 2.306\sqrt{\frac{4.461}{32.655}}$$

$$1.967 \pm 0.8523$$

$$(1.1147, 2.8193) \qquad \text{[2]}$$

Since 1.5 lies comfortably inside this confidence interval, then there is no evidence at all against the hypothesis that $\beta = 1.5$ at α=0.05. [1]